Aura-Node

Bridging the safety gap for every vehicle spectrum

Team Name:

Vision.exe

- Hack-4-Viksit Bharat Grand Prize Winner
- India Innovates '26, 3rd ranked finalist
- SIH '25 National Top 10

Members Name and Affiliation:

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PROBLEM STATEMENT

Non Line of Sight "Blind Spot"

Visibility Collapse

- Optical sensors, Cameras/LiDAR fail in India's dense winter fog, rendering luxury ADAS useless and causing high-speed chain-reaction fatal pileups.

Heterogeneous Reality

- Over 70% of Indian traffic comprises two-wheelers.
- Standard car-based safety systems ignore the most vulnerable riders on the road.

Golden Hour Deficit

- 80% of severe fatalities occur when victims are hidden off-road. Reliance on bystanders delays dispatch, missing the critical Golden Hour.

SOLUTION

Aura-Node

The Approach

A retrofittable, clinical-grade safety module designed for autonomous collision avoidance and emergency intervention.

Preventative Fog Mesh:

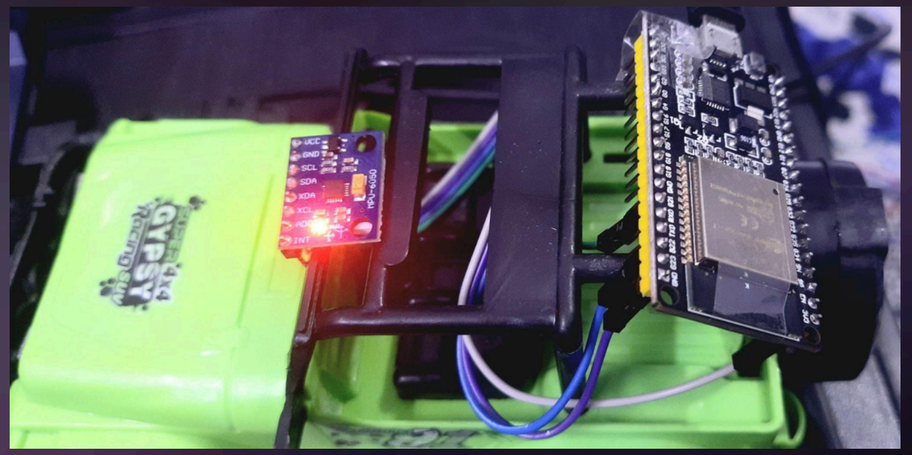
- Utilizes Non-Line-of-Sight (NLOS) ESP-NOW protocol to broadcast vehicle telemetry to local cars with ultra-low 2-5ms latency.

Golden Hour SOS:

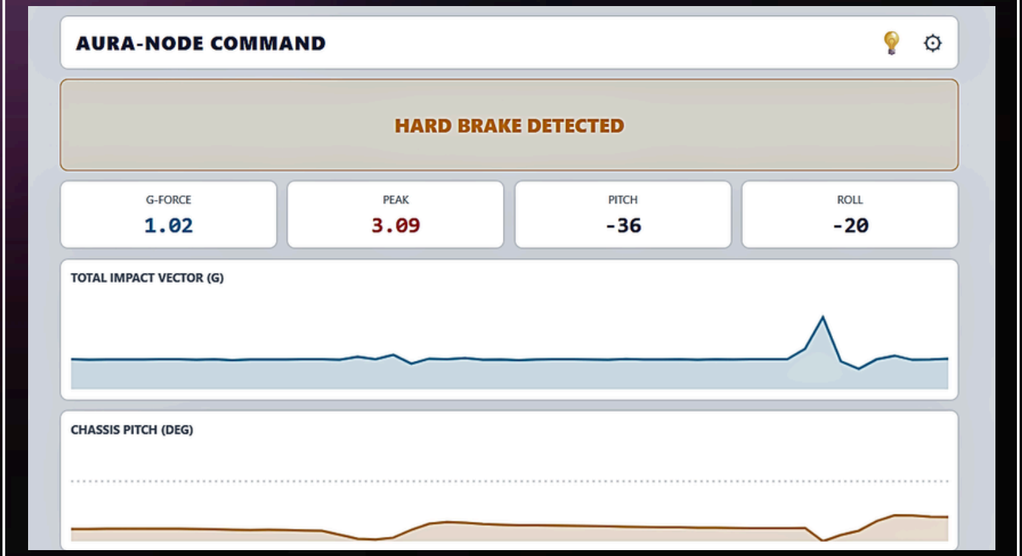
- An Edge-Kinematic engine continuously monitors 6-axis data to detect rollovers, instantly overriding the mesh to blast GPS coordinates.

Agnostic Retrofittability:

- Runs on a 3.7V LiPo or 12V Step-down. Protects a bicycle just as effectively as a commercial truck.

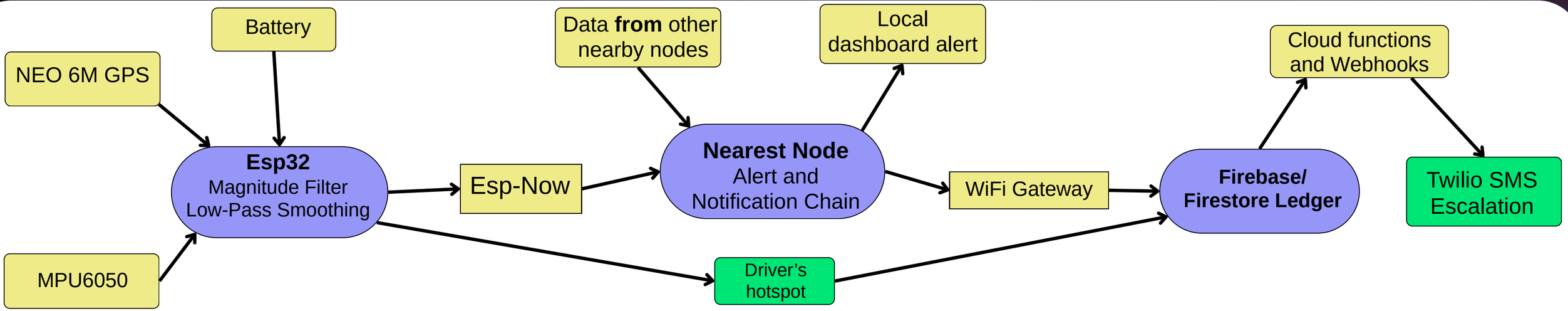


Edge Node Prototype



Prototype Dashboard

ARCHITECTURE



Perception Node

- **Hardware:** Integrated ESP32 Dual-Core MCU for parallel sensing and RF transmission.
- **Sensing:** MPU6050 6-Axis IMU for high-fidelity kinematics and NEO-6M GPS for exact localization.
- **Power:** Universally retrofittable. Runs on 12V vehicle step-down or a standalone 3.7V LiPo battery.

Intelligence

- **Vector Math:** Calculates the continuous absolute magnitude vector |A| at 100Hz directly on the edge.
- **Smart Filtering:** A low-pass software filter ignores noise and standard Indian potholes (spikes < 20ms).
- **Deterministic Trigger:** Executes a hard interrupt only on catastrophic impacts (>4G sustained) or a 90° Z-axis rollover.

Communication

- **V2V Mesh:** Utilizes the ESP-NOW protocol for zero-latency (2ms) local hazard broadcasting.
- **Data Mules:** Passes encrypted SOS payloads to nearby vehicles, which auto-upload to the cloud once they enter a 4G/5G zone.
- **Output:** Translates raw GPS telemetry into an automated Twilio SMS and Voice escalation to secure the "Golden Hour".

REAL-WORLD FEASIBILITY & MITIGATION

The Challenge

- **False Alarms:**

Indian roads have severe potholes. Won't the SOS trigger accidentally all the time?

- **Cellular Dead-Zones:**

Highway crashes often happen in areas with zero 4G/5G cellular coverage. How does the SOS get out?

- **Mass Adoption:**

V2V Mesh networks only work if everyone uses them. How do you get this on millions of vehicles?

Mitigation

- **Edge-Computing Kinematics:**

Our low-pass software filter and magnitude vector math (|A|) running at 100Hz on the ESP32 specifically isolates sustained catastrophic impacts from 20ms pothole spikes.

- **Data Mules & Store-and-Forward:**

If the crashed node and driver's phone have no internet, the encrypted SOS payload is securely passed via ESP-NOW to passing vehicles. Once those vehicles enter a 4G zone, they auto-upload the SOS to the Firebase Ledger.

- **Extreme Economic Scalability:**

By consolidating all logic and routing onto a single ESP32, our complete hardware Bill of Materials (BOM) is sub ₹1000. It requires no monthly SIM card subscription, making it a viable mass-adoption loss-leader for the government.

IMPACT / FUTURE SCOPE

National Impact and Features

- **Eliminating Pileups & Blindspots**

Bypassing optical sensor limits to prevent **chain-reaction pileups** in dense winter fog and sharp Himalayan blind-curves via **2ms NLOS mesh alerts**.

- **Universal Vehicle Inclusivity**

Extending clinical-grade collision avoidance to the **70% of Indian traffic (two-wheelers, rickshaws)** entirely ignored by luxury car ADAS.

- **Golden Hour Rescue**

Guaranteeing deterministic, **automated emergency dispatch** for hidden off-road crashes, significantly **increasing survival rates.**

Future Scope

- **Predictive Risk Profiling (AI)**

Aggregating long-term telemetry to map driver and vehicle behavior, **preemptively warning** the mesh if a **vehicle with high accident probability** enters the vicinity.

- **Smart City Maintenance**

Crowdsourcing **Z-axis anomalies (potholes)** across the mesh to provide municipalities with real-time, zero-cost **highway decay heatmaps.**

- **Commercial Cargo Safety**

Repurposing the edge gyroscope to profile heavy truck loads, **proactively warning drivers** of tip-over risks on **sharp highway curves.**

REFERENCES / LINKS

Github Repository

https://github.com/Priyanshunya/Aura-Node

Prototype Demo Video

https://drive.google.com/drive/folders/11mKqqmBnzBV_A0G_Bcpa5rW-wn8XLETZ?usp=sharing

Complete project Drive

https://drive.google.com/drive/folders/1zp2-XQhB8vsGtARqqAT5jZnl3VWJjKgX?usp=sharing

The Research Backbone

MoRTH India Road Accident Report:

WHO Road Traffic Injuries Fact Sheet:

Technical Framework

- **Espressif Systems (ESP-NOW)**
- ESP-IDF and Arduino Core documentation utilized to optimize the ESP-NOW peer-to-peer MAC routing and establish the ultra-low latency (2ms) V2V mesh topology.

https://docs.espressif.com/projects/esp-idf/en/latest/esp32/api-reference/network/esp_now.html

- **Kinematic Filtering (MPU6050)**
- Core libraries and mathematical frameworks used to train the Random Forest source apportionment models.

https://scikit-learn.org/stable/

- **Cloud & Escalation Architecture**
- Utilizes established environmental data science models to cross-reference particulate matter (PM2.5/PM10) signatures with ambient gas ratios (MQ135 data).

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8123863/